KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY

College of Engineering

ME 365: Thermodynamics II

FIRST SAMPLE PROBLEMS

SIMPLE RANKINE CYCLE

Answers are rounded up. They may even contain errors. Please report any error.

1. Water is the working fluid in a Carnot vapor power cycle. Saturated liquid enters the boiler at a pressure of 10 MPa, and enters the turbine as saturated vapor. The condenser pressure is 10 kPa. Determine (a) the thermal efficiency (ηth), (b) the work ratio, (c) the heat transfer to the working fluid per unit mass (qin) passing through the boiler in kJ/kg, (d) the heat transfer by the working fluid per unit mass (qout) passing through the condenser in kJ/kg.

Ans: (a) 45.4 %, (b) 63 %, (c) 1317.14 kJ/kg , (d) 719.19 kJ/kg

2. Water is the working fluid in a simple Ideal Rankine cycle. Saturated vapour leaves the boiler at a pressure of 10 MPa, and enters the turbine. The condenser pressure is 10 kPa. Saturated liquid enters the pump. Determine (a) the quality of the steam at the exit of the turbine, (b) the thermal efficiency of the cycle (ηth), (c) the work ratio, (d) the heat transfer to the working fluid per unit mass (qin) passing through the boiler in kJ/kg, (e) the heat transfer by the working fluid per unit mass (qout) passing through the condenser in kJ/kg.

Ans: (a) 66.19 %, (b) 37.22 %, (c) 98.94 % , (d) 2522.78 kJ/kg, (e) 1583.78 kJ/kg

3. A steam power plant operates on an ideal Rankine cycle. Superheated steam flows into the turbine at 2 MPa and 500°C with a flow rate ($\dot{m}$) of 100 kg/s and exits the condenser at 50°C as saturated water. Determine (a) the net power output ($\dot{w}_{net}$) in MW (b) the thermal efficiency ($\dot{\eta}_{th}$), and (c) the quality of steam at the turbine exit. (d) What would the efficiency be if the condenser temperature was lowered to 30°C?

Ans: a) 108.19 MW, (b) 33.22 %, (c) 91.26 %, (d) 36.5 %

4. Consider a steam power plant operating on the ideal Rankine cycle. The steam enters the turbine at 5 MPa, 450°C and saturated liquid exits the condenser at 10 kPa. The net power output (Wnet) of the cycle is 150 MW. Determine (a) the thermal efficiency (ηth) of the cycle. (b) the work ratio, (c) the mass flow rate (mdot) of steam, (d) the heat transfer (Qdot) in the boiler) the heat transfer (Qdot) from the condenser. (e) What would the thermal efficiency be if steam were superheated to 600°C instead of 450°C ? **Answers: (a) 82.25 % (b) 36.91%, (c) 99.56%, (d) 130.29 kg/s, (e) 3119.33 kJ/kg, (f) 1968.017 kJ/kg (g) 39.23%**

5. Steam is the working fluid in an Rankine cycle. Saturated vapor enters the turbine at 10 MPa and saturated liquid exits the condenser at 0.01 MPa. The net power output (Wdotnet) of the cycle is 150 MW. The turbine and the pump each have an isentropic efficiency of 85%. Determine (a) the thermal efficiency (ηth) of the cycle, (b) the mass flow rate (mdot) of steam, (c) the heat transfer (Qdot) into the working fluid as it passes through the boiler and (d) the heat transfer (Qdot) from the condenser to the steam as it passes through the condenser. **Answers: (a) 31.51%, (b) 188.79 kg/s, (c) 476.08 MW, (d) 326.08 MW**

6. Water is the working fluid in an ideal Rankine cycle. Superheated vapor enters the turbine at 15 MPa and 500 ^{o}C. The condenser pressure is 10 kPa. The turbine and the pump have isentropic efficiencies of 85 % and 75 %, respectively. Determine, for a flow of unit mass, (a) the thermal efficiency (ηth) of the cycle, (b) the net power output (Wdotnet), (c) the heat transfer (Qdot) into the working fluid as it passes through the boiler and (d) the heat transfer (Qdot) from the condenser to the steam as it passes through the condenser. (e) What would the net power output be if the mass flow rate of the working fluid were 100 kg/s?
Answers: (a) 34.98 %, (b) 1.08 MW, (c) 3.10 MW, (d) 2.01 MW, (e) 108.48 MW

RANKINE CYCLE WITH REHEAT

7. A steam power plant operates on a simple ideal Rankine cycle between the pressure limits of 4 MPa and 10 kPa. The boiler exit temperature is 600°C. Calculate:

(i) the quality of steam at turbine-exit

(ii) the net work output from the power plant

(iii)the thermal efficiency of the plant

The simple Rankine cycle is now modified to include a Reheat system. The reheat takes place at 1 MPa and the steam leaves the reheater at 600°C. Calculate:

(iv)the quality of steam at turbine-exit

(v) the net work output from the power plant

(vi)the thermal efficiency of the plant

Answers: (i) 89.58%; (ii) 1335.01 kJ/kg; (iii)38.38%; (iv) 98.38 %; (v)1620.76 kJ/kg; (vi) 40.77%

IDEAL RANKINE CYCLE WITH OPEN FEEDWATER HEATER

8. A steam power plant operates on an ideal regenerative Rankine cycle with a single open feedwater heater. The turbine inlet conditions are 10 MPa and 650°C, and the condenser pressure is 10 kPa. If bleeding takes place at 2 MPa, determine (a) the fraction of the steam that is bled (b) the enthalpy of the steam at the exit of the high pressure stage, (c) the quality of the steam at the exit of the low pressure stage, (d) the net work output, (e) the thermal efficiency (ηth) of the cycle, (f) the work ratio

Answers: (a) 23.85 %, (b) 3191.07 kJ/kg, (c) 85.20 % (d) 1277.21 kJ/kg, (e) 45.13% , (f) 99.11%

9. Consider a steam power plant operating on the ideal regenerative Rankine cycle with one open feedwater heater. Steam enters the turbine at 15 MPa and 600°C and is condensed in the condenser at a pressure of 10 kPa. Some steam leaves the turbine at a pressure of 1.5 MPa and enters the open feedwater heater. Determine (a) the fraction of the steam that is bled (b) the enthalpy of the steam at the exit of the high pressure stage, (c) the quality of the steam at the exit of the low pressure stage, (d) the net work output, (e) the thermal efficiency (ηth) of the cycle, (f) the work ratio

Answers: (a) 21.69 %, (b) 2820.30 kJ/kg, (c) 80.37% (d) 1297.62 kJ/kg, (e) 46.28% , (f) 98.72%

IDEAL RANKINE CYCLE WITH CLOSED FEEDWATER HEATER

10. A power plant operates on a regenerative vapor power cycle with one closed feedwater heater. Steam enters the first turbine stage at 10 MPa, 500°C and expands to 1 MPa, where some of the steam is extracted and diverted to a closed feedwater heater. Condensate exiting the feedwater heater as saturated liquid at 1 MPa passes through a trap into the condenser. The feedwater exits the heater at 10 MPa with a temperature of 175°C. The condenser pressure is 5 kPa. The mass flow rate ($\dot{m}$) into the first stage turbine is 270 kg/s. For isentropic processes in each turbine stage and the pump, determine (a) the mass flow rate ($\dot{m}$) of steam extracted from the turbine, (b) the thermal efficiency (η_{th}) of the cycle and (c) the net power developed ($\dot{W}_{net}$). **Answers: (a) 78.83 kg/s, (b) 42.44%, (c) 300.5 MW**

Watch out for the rest!!!